

Assignment: The Precision Prompting Challenge

Scenario

You are the newly appointed AI Strategy Lead at **AfyaTech**, a Nairobi-based health tech startup. Your team is building an SMS-based maternal health assistant for expectant mothers in rural Kenya and Uganda. Early testing shows your current AI prompts produce generic, urban-centric advice that ignores the following:

- Local dietary staples (e.g., matooke, ugali)
- Community health worker access
- M-Pesa payment constraints
- Cultural norms around pregnancy

Savannah Tracker Report

Introduction

Vague AI recommendations in maternal health can lead to deception, fear, or neglected therapy. Precise stimulation guarantees that recommendations take into account rural circumstances, such as mobile financial services, health care providers, commuting obstacles, and accessibility to food. AI transforms from an urban-biased advisor into a safe assistant by grounding suggestions in context and resources.

Nutritional Guidance & Rewritten Prompt (MAP + AIM)

A—Actor: Serve as a social nutritionist in rural areas of Uganda and Kenya, helping expectant women.

Input

users depend on traditional farming for survival and local markets. The majority of daily calories come from Matooke and Ugali. Low expenditures and firewood or charcoal are used for cooking.

M: Mission

Give weekly pregnancy meal recommendations that are reasonably priced and limited to locally sourced goods.

M: Memory The audience consists of low-income individuals who receive SMS counseling.

A: Resources include

legumes, beans, peanuts, milk, eggs, millet porridge, matooke, kale, ugali, kunde, managu, sweet potatoes, omena, avocado, and bananas.

P stands for prompt/action.

Create five everyday meal plans, snack options, and brief nutritional explanations in an easy-to-understand English/Swahili combination.

Improved sociocultural and economic alignment is the main improvement.

Prompt B: Reminders for Appointments

Rewritten Prompt (MAP + AIM)

A: Serve as an SMS reminder system for maternal clinics in rural East Africa.

I—Input

The majority of users may walk or take boda boda transportation because they live more than five km from clinics.

M: Mission

Boost attendance and cut down on missed prenatal care appointments.

M: Memory.

It takes one to three hours to travel. Clinics are only open on certain days. Community health workers assist with follow-up. M-Pesa is used by users to pay for transport.

A: Resources

Three days prior, one day prior, and the morning of the appointment are the times for reminders.

P stands for prompt/action.

Make three brief SMS reminders that promote CHW support and mobility planning.

Key improvement: Uses behavior-focused reminders to increase actual attendance.

Prompt C: Emergency Assessment

AIM + MAP + Verifier + Chain-of-Thought Rewritten Prompt

A—Actor: Serve as a maternity health triage assistant in rural Uganda and Kenya.

Input

Users may require calm, controlled direction and may not have access to rapid transportation.

M: Mission

Give advice about safe triage without inciting fear.

M: Memory

When symptoms are modest, safe home monitoring must be part of the advice.

Assets (Danger Signs):

strong headache, fever, convulsions, intense stomach pain, heavy bleeding, and no baby movement.

P stands for prompt/action.

Ask clarifying questions prior to offering advice (Verifier Pattern). Next, use a chain of thought to classify risks and propose calm next moves.

The main improvement is safer triage using phased questioning and argumentation.

I now see AI as a system that needs to be properly led rather than as a tool that provides answers thanks to this assignment. Without context, AI makes assumptions about cities and the world that could be harmful to individuals who are already at risk. I discovered through using AIM and MAP that incorporating local facts into prompts yields true value. The Verifier Pattern and Chain-of-Thought offer safety levels that are crucial in the medical field. AI is expanding the role of healthcare professionals, not replacing them. AI can enhance access to care in underprivileged places and provide moms with timely, culturally appropriate advice when it is based on practical limitations.

1. Failure of the MAP Framework: "Eat kale" in Kakamega
What exactly happened amiss with MAP: Memory Loss? No context for the location (Kakamega County)
No context for market access or income
Absence of a cultural diet setting

A: Loss of assets

There was a zero local cuisine database available.
Worldwide nutritional awareness → "kale" was AI's default.

P: Quick failure

The prompt was ambiguous ("nutrition tips") → AI used Western presumptions to fill in the blanks.

Outcome:
Rather than rural/local guidance, AI generated urban/global advice.

MAP Prompt Redesign (Localized) Contextual Memory:

Pregnant women in Kakamega County with modest incomes who purchase food from farms' nearby markets are the target audience.

Resources (local food reality):

Sukuma wiki, managu, kunde, terere, beans, maize, groundnuts, eggs, milk, sweet potatoes, omena, bananas, and avocados are among the food products that are abundant.

Act as a community nutritionist in Kenya as soon as possible.

Utilize primarily foods commonly found in Kakamega markets to provide advice on an iron-rich pregnancy diet. Provide instances of inexpensive meals.

2. Verification Method: Avoid panic triage.
Issue: AI immediately went to emergency advice. Redesign Prompt with Verifier Pattern

Serve as an East African rural maternity health triage helper.

Consider asking the user the following questions before offering advice: What is the number of weeks you are expecting?

Which complaints are you experiencing?

Is it mild, moderate, or severe?

How much time has it been going on?

Do you have access to a community health worker or transportation?

You should only offer advice after you have received responses.

Adopt proper home monitoring procedures if symptoms are modest.

3. The WHO summary is inferior to the Chain-of-Thought competitor strategy because WHO guidelines should be presented solely as information.
Your method, known as "chain-of-thought," involves making AI use step-by-step

reasoning. What does the WHO suggest? What are the guidelines' underlying assumptions, and which presumptions are incorrect in East African rural areas?

Adjust instructions to local limitations. Why this adds value: WHO believes clinic access is simple.

Transportation and financial obstacles exist in rural locations.

Local adaptation equals real-world execution, fostering genuine usability and trust. You go from information to practical advice.

How the **OCEAN Framework** helped me catch this before sharing it with donors?

4. Using OCEAN to identify out-of-date medical data:

1.8 doctors per 10,000, according to AI.

Utilized the OCEAN Framework: O—original, frequently repeated statistic→ probably recycled→ dubious.

C: Concrete

red flag: no year or source provided.

E: Clear chain.

All datasets and reports can be verified.

A: Self-assured.

There is a risk of hallucinations because the statement is certain and unambiguous.

N: Coherence of the narrative

Conflicts with known shortages in the healthcare workforce are irregular.

The outcome, OCEAN, prevents false information from reaching contributors by requiring authentication before distribution.